Statistical Foundations of Machine Learning
Practice Set 1 – Solutions

Problem 1

Question. Alice and Bob are among 8 students. Three are selected at random. Let A = Alice selected, B = Bob selected.

(a) Find $P(A)$. (b) Find $P(A|B)$. (c) Find $P(A \cup B)$.

Total outcomes:

$$|\Omega| = \binom{8}{3} = 56$$

(a)

Fix Alice; choose 2 from 7:

$$|A| = \binom{7}{2} = 21$$

$$P(A) = \frac{21}{56} = \frac{3}{8}$$

(b)

Given B:

$$|B| = \binom{7}{2} = 21$$

For $A \cap B$:

$$\binom{6}{1} = 6$$

$$P(A|B) = \frac{6}{21} = \frac{2}{7}$$

(c)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A) = P(B) = \frac{3}{8}, \quad P(A \cap B) = \frac{6}{56} = \frac{3}{28}$$

$$P(A \cup B) = \frac{9}{14}$$

Problem 2

Question. A fair coin is tossed three times. A = all same; B = at least one H and one T; C = at most one H.

Sample space has 8 equally likely outcomes.

$$P(A) = \frac{2}{8} = \frac{1}{4}, \quad P(B) = \frac{6}{8} = \frac{3}{4}, \quad P(C) = \frac{4}{8} = \frac{1}{2}$$

(a) A and B

$$A \cap B = \emptyset \Rightarrow P(A \cap B) = 0$$
$$P(A)P(B) = \frac{3}{16}$$

Not independent.

(b) B and C

$$B \cap C = \{HTT, THT, TTH\} \Rightarrow P(B \cap C) = \frac{3}{8}$$

$$P(B)P(C) = \frac{3}{8}$$

Independent.

(c) A and C

$$A \cap C = \{TTT\} \Rightarrow P(A \cap C) = \frac{1}{8}$$

$$P(A)P(C) = \frac{1}{8}$$

Independent.

Problem 3

Question. If A, B independent, show A^c, B^c independent.
Given:

$$P(A \cap B) = P(A)P(B)$$

$$P(A^c \cap B^c) = 1 - P(A \cup B)$$

$$= 1 - [P(A) + P(B) - P(A)P(B)]$$

$$= 1 - P(A) - P(B) + P(A)P(B)$$

$$P(A^c)P(B^c) = (1 - P(A))(1 - P(B))$$

$$= 1 - P(A) - P(B) + P(A)P(B)$$

Thus,

$$P(A^c \cap B^c) = P(A^c)P(B^c)$$

Problem 4

Question. Give examples where (a) $P(A \cap B) > P(A)P(B)$ (b) $P(A \cap B) < P(A)P(B)$
Experiment: one fair coin toss. $\Omega = \{H, T\}$.

(a)

Let $A = \{H\}, B = \{H\}$.

$$P(A) = P(B) = \frac{1}{2}, \quad P(A \cap B) = \frac{1}{2}$$

$$P(A)P(B) = \frac{1}{4}$$

Hence $\frac{1}{2} > \frac{1}{4}$.

(b)

Let $A = \{H\}, B = \{T\}$.

$$P(A) = P(B) = \frac{1}{2}, \quad P(A \cap B) = 0$$

$$P(A)P(B) = \frac{1}{4}$$

Hence $0 < \frac{1}{4}$.

Problem 5

Question. A and B alternately throw two dice. A wins on sum 9; B on sum 4. A starts. Find probability B wins.

$$p_A = \frac{4}{36} = \frac{1}{9}, \quad p_B = \frac{3}{36} = \frac{1}{12}$$

Let P be probability B wins.

$$P = (1 - p_A)p_B + (1 - p_A)(1 - p_B)P$$

$$P = \frac{8}{9} \cdot \frac{1}{12} + \frac{8}{9} \cdot \frac{11}{12} P$$

$$P = \frac{2}{27} + \frac{22}{27} P$$

$$\frac{5}{27} P = \frac{2}{27} \Rightarrow P = \frac{2}{5}$$

$$\boxed{P(B \text{ wins}) = \frac{2}{5}}$$